Chapter 16

Carbon Dioxide Snow Cleaning

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Chapter Outline

16.1 INTRODUCTION

Carbon dioxide (CO_2) snow cleaning is a mature, well-developed, and straightforward surface cleaning process. Its capabilities are well known and have been discussed previously; recent in-depth articles were published by Sherman.[1] Basically, a stream of small dry-ice particles strikes and cleans a surface via

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physical and solvent interactions. Particles of all sizes can be removed, from as small as several nanometers size to visible size. At the same time, CO_2 snow cleaning removes hydrocarbon and organic residues as well as water and solvent stains. The combined capabilities of particle removal over such a wide range and organic contaminant removal make CO_2 snow cleaning a process with wide applications.

Carbon dioxide cleaning is fast, gentle, and environmentally safe; it can satisfy the critical cleaning needs in research and industry. Furthermore, CO_2 is non-flammable and non-ozone depleting, and poses minimal risk for humans except for CO_2 buildup, frostbite (if applied directly to the skin), or oxygen replacement. The cleaning process is residue-free, non-destructive and carries away the contaminants for venting or capture. The fact that CO_2 is a dry process offers an advantage over many wet methods.

A key point is that CO_2 snow cleaning removes only contamination physically adhered to a surface; any contaminant that has reacted with the substrate, or is bound to the surface by ionic, covalent, metallic or other strong forces, will likely remain on the surface. A simple surface science view is that chemisorbed species remain; physisorbed species can be removed.

Here we will briefly review CO_2 snow cleaning processes, discuss equipment, applications, and finally compare CO_2 snow cleaning to other cleaning methods. The emphasis will be on original published sources and recent work since Ref. 1.

16.2 CO_2 SNOW CLEANING—BACKGROUND

16.2.1 Initial Investigations

First, we mention some initial investigations that established the history and effectiveness of CO_2 snow cleaning. Hoenig[2] was the first to publish an article, presenting only qualitative particle removal data. Whitlock[3] published the first set of controlled laboratory measurements that quantified the effectiveness of snow cleaning for micrometer and submicrometer particle removal. He reported a removal efficiency of over 99.9% for particles larger than 0.1 μm and the first hydrocarbon removal data. This work was expanded by Sherman and Whitlock,[4] who presented X-ray photoelectron spectroscopy (XPS) data quantifying hydrocarbon and organic removal from new and stained silicon wafers before and after cleaning.

16.2.2 Thermodynamics

A previous publication discussed the thermodynamics of CO_2 snow cleaning.[1] The key to understanding dry-ice formation and organic cleaning lies in the discussion on the CO_2 pressure-enthalpy diagram in that reference. Formation of CO_2 snow requires a high-pressure source, typically a cylinder at or 55 bar

(5.5 MPa, 830 psi). As the CO_2 passes through an orifice, the pressure and temperature drop, thereby forming a mixture of liquid and gas phases. When the pressure falls below 5.5 bar (0.55 MPa, 78 psi), any remaining liquid transforms to the solid phase, which is accelerated towards the surface by the untransformed gas. The snow percentage is determined by the source (liquid vs. gas) and the relative adiabatic nature of the transition. An adiabatic transition always yields a higher snow percentage. Mechanical interactions between the solid snow and surface contamination lead to cleaning and the CO_2 gas carries away the dislodged contamination.

Generally, nozzles used within the CO_2 snow cleaning industry are non-adiabatic, though one manufacturer makes an adiabatic nozzle. The adiabatic transition (constant enthalpy) allows for both liquid and gas CO_2 feeds. Non-adiabatic nozzles require a liquid CO_2 feed. The advantage of an adiabatic nozzle is a higher transition percentage to the solid phase for initial conditions and, more than likely, a higher velocity.

Two notable points are pertinent here. First, the term snow refers to not fully dense dry ice. This means that CO_2 snow lacks the impact energy compared to similar-sized CO_2 pellets used in dry-ice blasting. Second, this is a non-equilibrium process and using equilibrium phase and pressure–enthalpy diagrams serves only as a guide. Stating that the process goes through the triple point ignores the non-equilibrium nature of the process.

16.2.3 Cleaning Mechanisms

The above discussion focused on how to generate high-velocity snow. Next, we discuss cleaning mechanisms, which have been covered in detail previously.[1] Whitlock[3] was the first to publish the cleaning mechanisms for particle and hydrocarbon removal. Particle removal is based upon the flowing gas aerodynamic drag force and momentum transfer between the incident CO_2 snow and the surface particulate. For hydrocarbon removal, solvency is the dominant mechanism; for non-hydrocarbon organics, freeze fracture is the dominant removal mechanism.[5]

Particle removal is accomplished by the combined action of momentum transfer and the aerodynamic drag force. The drag force due to gas flow alone cannot generate sufficient force to remove micrometer- and submicrometer-sized particles bound by physical means such as van der Waals forces. Additionally, the aerodynamic drag force at or near a surface in the "dead zone" is zero. CO_2 snow cleaning introduces solid snow particles into the stream that strike the surface contaminant. These collisions transfer momentum from the snow to the surface contaminant, thereby potentially overcoming the weak binding forces and releasing the particle. Once released, the particle is carried away by the high-velocity gas.

Hydrocarbon removal requires liquid CO_2, which is an excellent solvent for non-polar hydrocarbons. During the short impact time, stresses increase at the snow-surface interface and the pressure can easily exceed the dry-ice triple point pressure of about 5.5 bar (0.55 MPa, 78 psi), yielding liquid CO_2 on the surface. When the snow rebounds off the surface, the pressure decreases and the dry-ice particle re-solidifies, carrying away the contamination. Whitlock was the first to publish this removal mechanism.[3] It must be emphasized that the distance between the nozzle and sample has a role in hydrocarbon removal since the snow velocity must be high enough to lead to the pressure rise above 5.5 bar (0.55 MPa, 78 psi) on the surface. Generally, depending on the high-velocity nozzles the author is familiar with, the distance can be as short as 0 cm (near contact) to 15 cm or greater.

Organics, not soluble in liquid CO_2, can be removed by a "freeze-fracture" process first discussed by Hills.[5] Generally, organics that are easily dissolved in liquid CO_2 are removed quickly. Organics with complex or other functional non-hydrocarbon-based groups are not readily soluble in liquid CO_2 and are removed slowly by "freeze-fracture." Here, it is believed that the snow freezes the deposit and breaks it off from the surface. In a publication discussed later, multilayers of siloxane were removed leaving just one chemisorbed monolayer. In addition, solvent stains and water spots can be removed with the freeze-fracture mechanism. This cleaning process is slower than for particles and hydrocarbons. Note that this makes CO_2 snow cleaning a method to clean spots after solvent cleaning or other wet cleaning processes. Figure 16.1 shows an example of a heavily stained optic that was cleaned on one side.

CO_2 snow cleaning is usually considered a high-velocity cleaning process. However, cleaning can occur at lower velocities with larger snow particles. Hill[6] studied the low-velocity snow cleaning mechanisms. Here the snow stream has a large expansion zone after the initial orifice, which allows the snow to agglomerate. As the snow increases in size, its velocity decreases. Hill[6] suggests the macroscopic snow does not contact the surface; instead, it glides above the surface and generates shear forces, resulting in particle removal. The released particle is "attracted and held to the snow flake by thermophoretic forces—a force arising from thermal gradients." This cleaning method is very

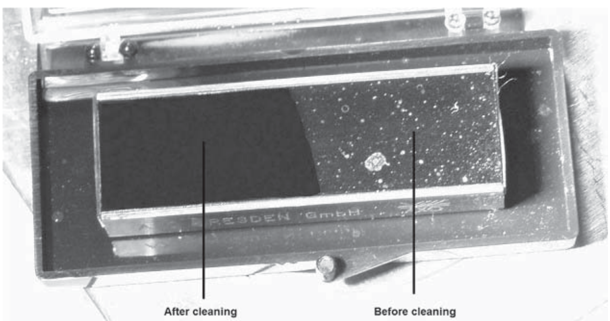

FIGURE 16.1 Example of CO_2 cleaning of a water-stained optic.

common for large telescope mirrors and is discussed later. The large snowflake generation requires "snow agglomeration" and this concept has been discussed by Liu et al.[7] This mode is not suitable for hydrocarbon removal unless there is a velocity boost.

16.3 EQUIPMENT AND PROCESS CONTROL

CO_2 snow cleaning systems are simple and straightforward. A basic system consists of a CO_2 source, the means to transport the CO_2 from the source to an on/off control, and a nozzle. On/off controls include valves (solenoids, pneumatic, and manual) and handguns. A typical manual unit is shown in Figure 16.2 with a pressure gage and an in-line filter. Costs start below \$2000. In the following sections, we will discuss some aspects of CO_2 cleaning equipment, methods, major cleaning issues, and process parameters.

16.3.1 Nozzles

The range of nozzles used in CO_2 snow cleaning is extensive, from simple orifices with expansion zones to complex multiple expansion geometries. Basically, the nozzle should allow for a controlled and directed CO_2 snow stream at the needed velocity. Sherman discussed previously nozzle options and the advantages of different approaches.[1] Nozzles are usually non-adiabatic, though adiabatic (Venturi or Laval geometry) nozzles are available, as originally published by Whitlock et al.[8] (double expansion) and by Sherman et al. (single expansion).[9] Adiabatic nozzles function with liquid or gas feeds. Non-adiabatic

FIGURE 16.2 A typical manual-based CO_2 snow cleaning unit with a hose, handgun, filter, and a nozzle.

nozzles from Hoenig and Hughes Aircraft and others date back to the same era (late 1980s) and require a liquid CO_2 feed.

Overall, all nozzles can clean by producing high-velocity streams of small dry ice. The snow size and velocity are determined by the orifice and expansion zone diameters. All nozzles have a small orifice before a larger diameter expansion zone. The adiabatic nozzle is characterized by a short conical nose cone expansion zone while the non-adiabatic nozzles use a longer cylindrical expansion zone. The expansion zone allows for the agglomeration and growth of snow at the expense of velocity and, at the same time, it focuses the snow stream. If no expansion zone existed, the stream would not be as good. This was discussed by Liu et al.,[7] who explored some of the geometric restrictions on the expansion zone and presented recommendations.

Other nozzle concepts have been developed. A simple tube nozzle surrounded by a much larger diameter expansion tube produces large snowflakes at a lower velocity. In order to compensate for the decreased velocity, several groups have introduced accelerating gases to increase the snow velocity. This produces higher velocities for the larger snow. Examples can be found by Swain et al.,[10] Jackson,[11] Goenka,[12] and others. These nozzles offer a range of dry-ice sizes and velocities, but they introduce added cost and complexity.

The large array of nozzles has led to particle removal in the nanometer range as verified by improved measurement methods. Whitlock's first measurements[3] in the late 1980s established the lower limit of 0.1 μm (100 nm). Since then, Jacobs,[13] using atomic force microscopy (AFM), and van der Donck et al.,[14] using scanning electron microscopy (SEM), reported cleaning down to 0.03 μm (30 nm); and recently Kim et al.[15] reported particle removal down to 0.01 μm (10 nm). Bowers et al.[16] have introduced a nozzle that has lowered this size limit further. Bowers et al. suggest that sizing the snow to the particle size allows the best energy transfer for dislodging the contaminant. This can be done with different nozzle designs and CO_2 feed phase. Incident CO_2 snow that will remove 10 nm size particles may not budge particles 500 times larger and the snow that removes the larger particles may not make the correct contact angle to remove the smaller particles.[16]

There are suggestions that smaller unknown organic particles can be removed as recorded by AFM, suggesting a lower limit of 0.003-0.005 μm (3-5 nm), discussed later in Section 16.4.2. It appears to this author that the challenge in nanometer particle removal is related to the cleaning method and metrology. In other words, the challenge is both the cleaning and measuring method and verifying the same or similar areas before and after cleaning.

The nozzles described above are point sources and clean areas smaller than one or a few centimeters diameter. Layden[17] was the first to develop a unique large-area nozzle. It offered a double expansion nozzle whose exit can continuously range in width from 1 to 10 cm and can operate with either gas or liquid CO_2. Krone-Schmidt[18] and later Askin and Bowers[19] and Yoon[20] showed special multiple orifice large-area nozzles or manifolds composed of individual nozzles. These nozzles operate as point sources assembled into an array, and thus may lack a continuous exit stream.

16.3.2 Equipment Needs and Cleaning Issues

As discussed above, CO_2 snow cleaning equipment can be simple and nozzles are the key. The remainder of this section discusses process issues—in other words, the methods and equipment needed to insure proper cleaning. CO_2 snow cleaning requires attention to sample support, means to counter surface cooling, electrostatic charge, and ways to avoid recontamination or new contamination. Sample support depends on each sample and is not discussed. Moisture condensation and recontamination risks are major process parameters that require attention. To counter these issues, process controls and equipment have been developed and have been discussed before.[1,21]

16.3.2.1 Surface Cooling

Dry ice is cold, about 194 K (−79 °C). The high velocity of incident snow will cool most surfaces quickly, instantly for most insulators. With proper movement, the surfaces never got this cold, and most surfaces stay above 0°C. This causes moisture condensation on the sample surface, which can potentially interfere with cleaning. Therefore, the user must minimize or eliminate moisture condensation. Simple methods include heating with a hot plate, infrared (IR) lamps, hot-air guns,[3,4] or using heated or unheated nitrogen purges.[1,11,22] Other users have used dry boxes with and without load locks. The best dry boxes operate with a dew point in the range 253-223 K (−20 to −50 °C) and have load locks, along with recirculating dry atmospheres and HEPA filters.

Cherry[23] and Krone-Schmidt and Markle[24] were the first to develop and patent dry boxes, and many more have been developed over the years for specific applications. These types of dry boxes have worked well for small and mid-size samples such as wafers, hard drives, and optics.[25] For larger samples, this approach will fail because the load lock will take too long to dry out after changing samples. DePalma and Sherman[26] introduced an alternative method to the sealed dry-box and load-lock assembly. They described a system with a high-flow dehumidifier in the main chamber, thus avoiding any external load lock. A 20-m^3 chamber was dried to a dew point below 243 K (−30 °C) within 4 min using this approach.

It should be noted that surface cooling is much less a problem with the low-velocity mode commonly used for telescope cleaning.

16.3.2.2 Static Control

Electrostatic charging can be a problem, as shown by Hills.[27] Antistatic devices are also used at times while cleaning. Commonly, grounding a sample is a good start, but insulators require more attention. Common antistatic discharge sources can be placed near the sample or even placed on the gun to assist in static control. Generally, antistatic discharge bars are placed in dry boxes to assist in these dry environments. Bowers,[28] Krone-Schmidt et al.,[29] and others[11] have described various devices. Common antistatic sources and guns make good accessories to any CO_2 snow cleaning system.

16.3.2.3 CO_2 Purity

Impurities within the CO_2 source are a much greater issue for precision cleaning. Whitlock,[3] Hill,[6] and Sherman et al.[9] performed early investigations on the sources and consequences of these impurities. Whitlock,[3] in the mid-1980s, identified a source of submicrometer particulate residue as heavy hydrocarbons, possibly lubricants used during the purification process. Under best conditions, this residue was less than one submicrometer particle per square centimeter. Hill[6] identified polymeric residues on a surface after CO_2 snow cleaning and traced them back to hose materials. Sherman et al.[9] compared different CO_2 purity sources to clean a "clean" surface. They identified the supercritical CO_2 grades (without the He headspace) as the best CO_2 source. Liquid-fed sources led to greater contamination than similar gas-fed sources; since liquid CO_2 is an excellent solvent, it is expected that the hydrocarbon contamination in a cylinder is concentrated in the liquid.

For large-scale applications, on-site purifiers from manufacturers and industrial gas companies can reduce the heavy hydrocarbon contamination by a factor of 10-100 or more.[30] Commercial purifiers are really condensers using a CO_2 gas feed that yield a liquid CO_2 source. Other methods for CO_2 purification include specialized catalysts[31,32] or in-line sieve products.[33]

16.3.2.4 Equipment

Contamination sources can also come from the equipment. The equipment can shed particles, and these particles can enter the CO_2 feed stream. Therefore, care is required in choosing material and surface finishes. For critical cleaning applications, electropolished stainless steel tubing and valves are best suited along with parts cleaned to industry standards. Point-of-use filtration is strongly recommended. This means placing a high-quality filter just before the nozzle, if possible, and filter specifications should be the best possible for the process.

16.3.2.5 Methods and Work Space

Once a particle or organic contaminant is removed from the surface, it must not redeposit on the cleaned areas. This can happen from improper workspace areas and also from poorly chosen cleaning techniques. A small cleaning space allows the CO_2 streams to strike side walls and other items and create a return flow[6]; a large space is best.

The cleaning procedure is also critical. Generally, cleaning should proceed from one side to the other to ensure that the stream cannot flow back to the surface or over the cleaned region. Cleaning must proceed from a cleaned area into a dirty area. Cleaning must never be performed toward previously cleaned areas; the stream must always be aimed at still-to-clean areas and toward a vent. For critical cleaning, it is essential that the exhaust stream be removed from the cleaning area, before venting or filtering.

16.3.2.6 Surface Damage

A common question is whether the incident dry-ice snow can damage a surface. In almost every case, the answer is no. In nearly all industrial and the vast majority of research and high-technology applications, CO_2 snow cleaning is a damage-free process. Yet there are limitations within the process, mostly related to severe thermal shock, sample geometry, and other factors. Our experience suggests that only the rare sample has problems. If a sample is properly supported, it can be cleaned and even small flexible items such as individual fiber-optic connectors have their ends cleaned. Optical and electron microscopies have failed to find damage on most surfaces even at high magnifications. Patterned wafers can be cleaned,[4,14] while microelectromechanical systems devices pose challenges, through they have been cleaned under certain conditions.[34] One user of fiber optics successfully cleaned 1-mm fiber (sticking out up to 1 cm) and 500-μm fiber (sticking out ~5 mm).[35] AFM[36–38] has shown lack of surface damage, as discussed later. Another example can be found in Brandt and Simpson,[39] who analyzed the smoothness of Au on glass before and after CO_2 snow cleaning. The AFM data showed a smoothing effect that was not evident in reflectivity measurements, supporting minimal atomic movement and no abrasion.

16.4 APPLICATIONS

CO_2 snow cleaning is versatile. Cleaning applications can span from simple laboratory applications to production contamination control problems. These applications include cleaning many different materials, optics, vacuum components, hard-disk-drive components, process tools, wafers, semiconductor tooling, photomasks, art conservation, and much more. Many patents exist along with numerous unpublished reports on cleaning applications developed by manufacturers and users. Several of the CO_2 snow cleaning equipment manufacturers have developed applications as part of their work and have released the data and applications to the public. As a result, the technique has gained wider acceptance and has entered many new fields. We will discuss a few applications and patents here to indicate the wide range of uses and capabilities.

16.4.1 Materials

In the early 1990s, Sherman et al.[9,40] published data demonstrating particle and organic contaminant removal using CO_2 snow cleaning from a wide variety of materials, including wafers, metals, ceramics, glass, optics, coated optics, mirrors, patterned wafers, and polymers. Samples were characterized before and after cleaning by surface analysis and by optical or electron microscopy. Essentially, if one can support a sample to withstand the incident snow stream, cleaning is possible.

16.4.2 Analytical Sciences and Surface Science

In the past, Sherman and coworkers have shown CO_2 snow cleaning to be a powerful tool for many surface analytical methods such as XPS.[9,40] The chemistry of the top dozen atomic layers is analyzed by these methods and clean samples are obviously critical. By similar reasoning, this extends to other methods such as Auger, secondary ion mass spectroscopy,[41] AFM, and other methods. Cleaning is not limited to standards and samples, but the electron or ion optics and vacuum equipment also can be cleaned.[40–44] CO_2 snow cleaning has been used on SEM samples, but is not common since air blow-off and *in situ* quasi-plasma cleaning methods exist.

AFM sample preparation using CO_2 snow is now common, from the first comments by Morris[36] to the recent paper by Chernoff and Sherman[38] in 2010. The latter paper showed that CO_2 snow cleaning leads to better imaging, cleaner surfaces, and restoring old step-height standards to new condition, i.e. the same step heights as “new”. Figure 16.3 shows a heavily contaminated step-height standard in both height and phase contrast before CO_2 snow cleaning. Figure 16.4 shows the results from a similar region after CO_2 snow cleaning. Cleaning removed over 99.5% of the contamination. No substrate material was removed as the step heights were unchanged from the initial values. The AFM data in Figure 16.3 shows a typical contaminant height of about

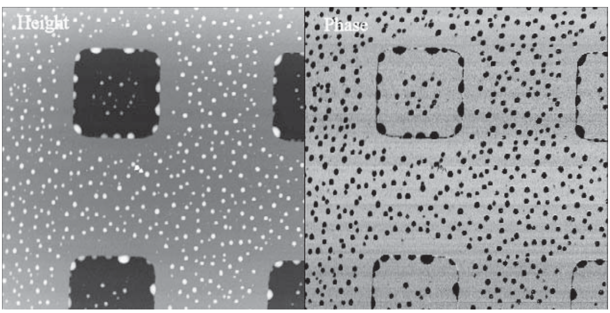

FIGURE 16.3 Height and phase image of the contaminated surface before cleaning of an 18-nm step grid (15-μm scan).

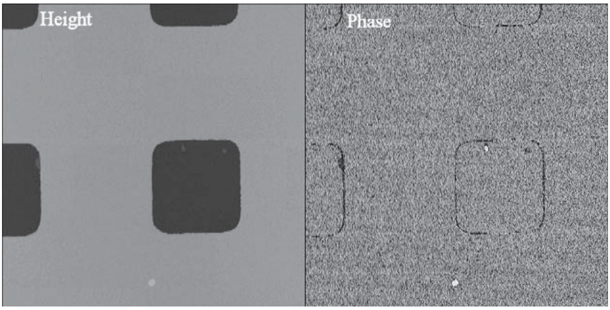

FIGURE 16.4 Height and phase image of the contaminated surface after CO_2 snow cleaning of an 18-nm step grid (15-μm scan).

5-25 nm. After cleaning, few particles remained of the unknown contaminant (probably organic). Since all particles larger than 5 nm were removed along with 96% of the particles 3-5 nm high, it appears that a lower limit for CO_2 snow cleaning is below 5 nm for hydrocarbons and organics. In the future, it may be shown that 5 nm is the limit for inorganics as well. Particles less than 3 nm in height were excluded from the automated analyzes due to surface roughness and noise issues. Details are provided in the paper by Chernoff and Sherman.[38]

Nanoprobe indent tip cleaning with CO_2 snow has been discussed by Morris[45] and he found it to be more effective than solvent cleaning.

A recent paper by Hugall et al.[46] showed that CO_2 snow cleaning can vastly improve the signal-to-background ratio on surface enhanced Raman scattering and also enhance the signal. The reasons for the improved signals are not fully understood, but the authors suggest that CO_2 snow cleaning “results in massive improvements in signal.” Details are provided in Ref. 46. CO_2 snow cleaning has been used by other optical spectroscopy techniques. One example is ellipsometry,[47] where CO_2 snow cleaning of SiO_2 on Si standards led to more stable surfaces over time.

Other papers within surface science point out unique results when producing clean surfaces. One example is reported by Chow et al.,[48] who showed CO_2 snow cleaning can lead to “perfect self-limited siloxane monolayers.” It seems that cleaning removed defects, siloxane overlayers, and the experimenters were left with the desired self-assembled monolayer. This result may yield new ways to deposit one-monolayer-thick films that are common in many chemical studies. Additionally, the result is in line with the expectation that CO_2 snow cleaning only removes physically adsorbed contamination and not the chemically bound first monolayer. Similar results were found by Bhatt et al.[49] Other have used CO_2 snow cleaning not only to clean the substrate but to make unique thin films after deposition for nanotube circuits. Quoting Professor Lay,[50] “the snow jet was used to clean the surface and remove excess silane. The silane, which acts as an adhesion layer, polymerizes, forming multiple weakly adsorbed layers. The snow jet is the best way to removes the excess layers, while leaving behind the critical single monolayer." Nanotube circuits have been made after substrate cleaning with CO_2 snow. Snow et al.[51] used a photoresist process to protect the positioned nanotube before CO_2 snow cleaning. After cleaning, they removed the protective photoresist on the nanotube, yielding a nanotube as the gate on a field effect transistor.

16.4.3 Vacuum and Thin-Film Technologies

Many applications in vacuum technologies have been developed, including the cleaning of vacuum systems and parts, electron, ion, and X-ray optics, mass spectrometers, samples for surface analysis, and thin-film substrates. Previously, we discussed in detail the data generated by Layden and Wadlow,[42] demonstrating that cleaning of a residual gas analyzer by CO_2 snow was faster and more effective than solvent cleaning. Other similar cleaning examples have been found in many electron and ion guns in vacuum systems; a good example is by Bailey and Midavaine,[43] describing the use of CO_2 snow cleaning of electron guns.

One interesting application is the cleaning of the stainless steel surfaces during fabrication. Snow cleaning removes residual machining oils, particle residues, and other debris from the metal surfaces[52] and welding parts before welding.[53] This process has led to less staining near the weld zones and fewer weld defects. CO_2 snow cleaning has also been used to assist in cleaning vacuum parts (aluminum and stainless steel based) for stain removal after cleaning by other methods.[52]

A recent application in vacuum technology has been cleaning radiofrequency cavities for high-energy physics accelerators.[44] Cleaning reduced high-voltage discharge events with the removal of metallic stringers or field emission spots.[44,54,55]

Cleaning thin-film substrates before optical or any thin-film coating is quite common with CO_2 snow. A good recent example related to cleaning ITO (indium tin oxide) on solar cells was discussed by Wang et al.[56] Here improved yields on organic-based solar cells were found when compared to standard cleaning methods. This result is typical of many examples of substrate preparations before coating. In fact, this is quite a common application in thin-film coating. Large-area substrate cleaning for glass or reel-to-reel polymer coating presents some challenges due to the substrate size. Cleaning systems have been designed for both applications.[57,58]

16.4.4 Optics

Cleaning glass substrates before coating or during or after assembly has become a major application area for CO_2 snow cleaning. Cleaning can serve as either the initial or the final cleaning step for uncoated and coated glass, lenses, IR, ultraviolet (UV), fiber-optic ends and connectors, and ceramic or semiconductor substrates, or it can serve to clean individual optical components during production or assembly. There are many diverse uses, though the optics industry does not allow the publication of manufacturing or developmental details. The patent literature and its references provide hints of the details.[59] Many optical cleaning examples exist, ranging from both coated and uncoated lenses, mirrors, gratings, IR and UV, satellite optics, lasers, large telescopes, and much more. The key to proper optical cleaning is to insure a dry, static-free environment.

Diamond-turned metal optics are commonly cleaned with CO_2 snow as a means to remove oils and residual contamination. Although most users are from industry and do not publish details, one diamond-turned machining paper was published by Goss et al.,[60] where they machined GaAs wafers and used CO_2 snow to remove machining debris. These results are common for diamond-turned optics.[61]

16.4.4.1 Telescope

CO_2 snow cleaning has been used for over 25 years on large telescope mirrors. The cleaning systems are usually based on low-velocity CO_2 snow cleaning. Many of these facilities have made their own cleaning systems or have obtained units commercially[62] and published literature is available on this topic. Most systems are based on a single or multiple 2.5-cm exit tube, though Sherman recently introduced the first 5-cm-wide first slit nozzle.[63] Generally, what has been found is that snow cleaning can return a telescope mirror reflectivity to its initial values. For large mirrors, this is a rapid cleaning process. The author has seen an 8-m mirror cleaned in under 30 min.

16.4.5 Wafers, Semiconductor Processing, Hard-Disk Assemblies

In addition to particle removal from both the front and back sides of wafers,[64–66] many other applications have been developed. Examples include metal lift-off,[67,68] photomask cleaning and removal,[69,70] mask repair,[71] die removal,[72] cleanroom and process tooling cleaning,[73,74] post-CMP (chemical mechanical planarization) cleaning,[75] and more applications. The reader is directed to the patent literature and the Eco-Snow website (www.eco-snow.com). These systems require process control environments and are almost always fully automated systems with process-controlled mini-environment chambers. Wafer-carrying cassettes were cleaned using CO_2 snow in a dry environment with a conveyor, multiple orifice nozzles, and static control.[76] Wafer-handling equipment and even individual chips can be cleaned using CO_2 snow. In addition, many different types of chip modules have been cleaned before packaging.

Cleaning of hard-disk assemblies and components has always presented challenges to contamination control engineers and CO_2 snow cleaning has met these needs. Almost all parts of a typical hard-disk-drive assembly have been successfully cleaned. The key concern for this industry in cleaning parts is to insure there is no recontamination of the parts. Semi-automated and automated disk cleaning systems have been discussed.[77–80]

16.4.6 Other Applications

New or unique applications are wherever cleaning is needed beyond compressed air or simple solvents. We will summarize a few applications, with the first being the most promising and the last rather surprising.

16.4.6.1 Art and Artifact Cleaning

CO_2 snow cleaning has been used as a dry, non-contact cleaning method for art and historical artifacts with the aim of removing debris without damage. The lack of damage is of paramount importance here because the samples tend to be impossible to replace. The Smithsonian American Art Museum and the Arizona State Museum have developed cleaning methods and provide examples. Shockey[81], using a CO_2 gas source, cleaned polymer-based minimalist sculptures that had small white particulates on the surface. Additionally, a nitrogen shield was used to minimize moisture condensation. Zimmt et al.[82] and then Odegaard[83] looked into cleaning baskets and textiles for general cleaning and possible pesticide removal with CO_2 snow. Cleaning museum artifacts may be a growth area for CO_2 snow cleaning. The author has performed tests on many objects; many objects are not suitable for cleaning, or snow cleaning did not assist in removal. Old oil-paint-based paintings are not suitable for CO_2 snow cleaning due to potential cracking and flaking. One cleaning example, although not art restoration,[84] showed about 95% soot removal from a burnt kitchen item. Removal of soot deposits is a likely application of CO_2 snow cleaning in this field. These items were all discussed at a recent conference at the Smithsonian American Art Museum in September 2015. Details are available from the author.

16.4.6.2 Biofouling

Several researchers[85–87] have demonstrated effective 91-99% removal of *Escherichia coli* biofilms under several conditions, as well as immobilized proteins from a surface. This is the opposite of the findings by Buss et al.,[88] who failed to remove such a high percentage of *Bacillus* sp. with CO_2 snow cleaning. Users must test these applications first.

16.4.6.3 Cooling and Cleaning

Parsells et al.[89] used multiple CO_2 snow jets to clean and cool diamond saw blades that were used to cut up and decommission the Princeton Plasma Tokamak Fusion Test Reactor. Cooling is a common CO_2 snow application.

16.4.6.4 Additives

Solvents and other species have been mixed into the liquid CO_2 stream to create co-solvent-based cleaning. Jackson[11] mentioned co-solvent additions back in 1998 and expanded these ideas later.[90] Askin and Bowers[19] used a multiple-nozzle assembly that allows for "the addition of co-solvent cleaning agents to the CO_2 snow."

Another approach was introduced by Banerjee and Chung,[91] where they applied "at least one liquid and cryogenic (spray) simultaneously … or sequentially." The liquids included common solvents such as acetone. These authors later extended this concept to reactive and non-reactive gases including ozone, vaporized solvents, nitrous oxide, and other species.[92]

16.4.6.5 Plastic Parts

Keklik[93] of BMW (auto manufacturer) applied for a patent using multiple CO_2 streams to remove dust and debris from plastic automotive parts. The patent is not clear what form of CO_2 nozzles are in use, but it appears to be a snow jet. We believe they used a low-velocity mode and then added compressed air for increased velocity.

16.4.6.6 Bedbugs

As a final example, one never knows where applications might be found. One interesting area has been killing bedbugs, which requires a low-velocity CO_2 stream and a means to locate the critters (usually, sniffing dogs). Extermination is accomplished by freezing the vermin.[94] The author was quite surprised to see this work and the cleaning units can be quite simple.[95]

16.5 CLEANING COMPARISONS

Here, we compare CO_2 snow cleaning to other cleaning methods. The cleaning methods for comparison include water- and solvent-based methods, such as ultrasonics, wiping, blowing, vacuum methods, lasers, replica tape, and argon-nitrogen cryogenic aerosol. An excellent summary, though aimed at the optics industries, has been authored by Schalck and published by SPIE.[96] Finally, we compare CO_2 snow cleaning to other CO_2 cleaning methods.

All cleaning methods have their advantages and disadvantages, including capital and operating (labor and consumables) costs, time per sample, effective cleaning range, and cleaning efficiencies. It is imperative to note that no one method is perfect, they all have their limitations. Vendors have to be honest with customers, and if they fail to mention what can be an issue, that cleaning method should be avoided. (As an example, the author heard a replica tape vendor say no issues exist, and then was asked about incomplete removal of the replica tape, which happens.)

Aqueous and solvent-based cleaning methods, with and without ultrasonics, are batch methods, meaning one or a large number of samples can be cleaned simultaneously. This contrasts with CO_2 snow cleaning, which is an in-line method, meaning only one sample can be cleaned at a time per nozzle. The obvious advantage of these wet methods is volume production, and the main limitation is drying and potential staining. A manufacturer cleaning hundreds of 10-cm^2 optical glass samples a day will most likely choose ultrasonics due to the volume throughput and will have to insure proper drying to avoid stains and spots.

Manual wiping with solvents is usually quick, and common in the optics industry. The limitations are labor costs and drying issues. Yet, when labor is not an issue, it is acceptable and can remove particles and organics. Potential surface marring from wiping hard particles on soft optical surfaces can occur, and users must be aware of this limitation. Additionally, stains from improper drying can occur. This process can be rapid per part and the user can quickly inspect each sample after cleaning.

Air blow-off with compressed air or nitrogen is quite common and works well for particles larger than several micrometers. Cost per sample is quite low. The main concerns are oil impurities within the air compressors and lack of effective filtration. Air blow-off does not work on hydrocarbons or particles smaller than a few micrometers.

Vacuum methods include plasma, glow discharge, sputtering, etc. These methods work phenomenally well on many substrates and they all require a vacuum system. Plasma may be the most effective method for removal of organics, but potential substrate oxidation is a concern. Plasma is not effective for inorganics. UV-Ozone for organic removal can be done outside a vacuum system, but lacks any ability to remove inorganic particles.

Argon-nitrogen cryogenic aerosol cleaning is quite similar to CO_2 snow cleaning. It was first reported by McDermott back in 1991[97] and developed over the years. Essentially, it is a similar process in that a cryogenic particle is aimed at the surface and relies upon momentum transfer for cleaning. Many patents have been granted. Compared to CO_2 snow cleaning, results may be similar for particle removal and cleaning systems, with good results having been reported.[98]

There are a few replica-based cleaning companies that offer polymers in a solvent- or water-based solution. Cleaning is effective, but the drying times make the process slow in comparison to most all other techniques. Cost per sample can be quite high too. Issues exist, as Kohli stated: “the coating tends to cling to the irregular surface, loses its coherence, and breaks off in pieces when peeled."[99] A strong area for this process may be gratings (where damage by CO_2 snow cleaning is a risk if not done properly) or smooth lenses. Residual polymers have been noted by XPS studies and visual examination.[99,100] Again, as for all cleaning methods, testing is required before application.

Cost per part is an important consideration. Large automated cleaning systems are quite expensive and are not suitable for cleaning the occasional sample unless there are special considerations that justify the cost. Automated CO_2 snow cleaning systems costs are high, and their applications have been directed at high-value samples such as photomasks, back-end wafer processing, and unique high-value parts (gyroscopes). The equipment costs are justified in these cases, as they are for in-line ultrasonics or solvents, or any automated cleaning process. Laser particle removal systems fall in this category too because large capital expenses must be taken into consideration.

Other CO_2 cleaning methods include liquid and supercritical CO_2 cleaning, and dry-ice pellet blasting. Liquid CO_2 cleaning uses the excellent solvent properties of the liquid for degreasing and cleaning.[101] Adding surfactants or co-solvents has made liquid CO_2 suitable for textile dry cleaning or metal degreasing. Inorganic particle removal is a weakness here. Systems can be expensive. Supercritical CO_2 refers to liquid CO_2 exceeding the critical temperature (305 K or 32 °C) and pressure (74 bar or 7.4 MPa), where the liquid CO_2 loses its liquid characteristics and enters a unique phase that combines the best properties of both liquid and gas phases. Supercritical CO_2 has phenomenal solvent properties and the penetrating properties of a gas having zero surface energy. Applications abound in chemistry and biochemistry, along with a few semiconductor applications. The main limitations are no particle removal, and high temperature and pressure requirements make the systems more complex and costly.[101] Dry-ice pellet blasting uses compressed air to accelerate macroscopic pellets of dry ice. The dry-ice impact energies are much higher than with snow and can remove gross contamination in many industrial applications where CO_2 snow lacks the impact energy. The need for compressed air makes these systems more expensive.

Overall, all cleaning methods have their advantages and disadvantages and the user should expect the suppliers to be honest about the limitations.

16.6 SUMMARY AND CONCLUSIONS

Carbon dioxide snow cleaning has evolved into an effective surface cleaning process with an ever widening array of applications. The particle removal abilities of CO_2 snow from visible sizes down to the nanometer range, along with hydrocarbon and organic removal, are unique. Cleaning is accomplished without surface damage on most surfaces. We discussed many applications over a wide range to emphasize its capabilities and mentioned limitations and comparisons to other cleaning methods as well. It is imperative to understand the simplicity of the process and equipment prices can be less than $2000 for manual cleaning systems. Automated systems, though quite expensive, can address state-of-the-art cleaning applications within the semiconductor and other industries.

Overall, the work done in the past 25 years has taken CO_2 snow cleaning from a laboratory niche cleaning method to its present state as an accepted solvent-free, dry cleaning process with widespread applications.

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